1.63 Large Data Set: OCR A

Worked Solutions

1 a)

Work from home

Underground, metro, light rail, tram

Train

Bus, minibus or coach

Motorcycle, scooter or moped

Driving a car or van

Passenger in a car or van

Taxi

Bicycle

On foot

1 b)

Not in employment

1 c)

2001 and 2011

2)

New unitary authorities were made in 2009, meaning that when creating the data set for 2011 some of the data was summed to create the new unitary authorities.

3)

People who predominantly drive (which is likely to be the case for those using park and ride) will be in the "Driving a car or van" or "Passenger in a car or van" category.

4)

Not in employment as the Census was based on the week before the Census.

5)

Exeter has a big university which explains a noticeably larger number of people aged 20-24.

6 a)

$$\bar{x} = \frac{1915}{36} = 53.2$$

$$s_x = \sqrt{\frac{196261 - \frac{(1915)^2}{36}}{35}} = 51.93227427\ldots = 51.9$$

6 b)

He didn't include the Isles of Scilly which should have 0 in the large data set rather than the dash.

$$\bar{x} = \frac{1915}{37} = 51.8$$

$$s_x = \sqrt{\frac{196261 - \frac{(1915)^2}{37}}{36}} = 51.94730525\ldots = 51.9$$

7)

A baby less than 1 year old is categorised as 0 years old (as it is either based on age at last birthday for the 2001 data or their age on 27 March 2011 for the 2011 data.)

8 a)

Systematic sampling

8 b)

Quota sampling

8 c)

Stratified sampling

8 d)

Cluster sampling

9 a)

No. Whilst the number of 25-44 year olds has gone up, the percentage of this category has only gone from 35.3% to 35.5% so it has only marginally increased.

9 b)

As 10 years has passed the 5-14 years olds have gone into the 15-24 year old category, which has increased by 214 441. However, this doesn't account for people leaving the category such as moving to a different area.

9 c)

The category includes people over 94 and up to 110 and 115 for 2001 and 2011 respectively.

10 a)

This point is where a lower proportion of people drive to work but a higher proportion do not necessarily take public transport. This point is a rural area (it is Eden which is the Lake District) and demonstrates a high proportion of people who must work from home.

10 b)

45.15% will use another means of transport.

11)

On average, the proportions of people using bus, minibus or coach across the country hasn't changed. Both data sets have a (negative) skew but but the middle 50% of people using bus, minibus or coach across the country has gone down from 2001 to 2011. The range is similar, but by 2011 the proportions of people using bus, minibus or coach across the country has become more consistent.

12)

Category 5 will be other. London will be really high public transport use, rural will be significantly lower which would suggest category 4. Identifying A as London and B as the rural authority. Rural will be high car use, with London lower so 'car as a passenger or a driver' will be category 2 Category 1&3 are not significantly different enough, but will be the remaining categories which are working from home and bicycle or on foot, which suggests 1 as being working from home which was much more common in rural areas and 3 for on bicycle or by foot.

13 a)

fd	x	fx	fx^2
213	8.75	1863.75	16307.8125
215	21	4515	94815
229	34.5	7900.5	272567.25
229	52	11908	619216
173	67	11591	776597
90	82.5	7425	612562.5
1149		45203.25	2392065.5625

Mean is $\frac{45203.25}{1149} = 39.3$ years old (3 sf)

Std dev: $\sqrt{\frac{2392065.5625}{1149} - \left(\frac{45203.25}{1149}\right)^2} = 23.1$ (3 SF)

13 b)

$$6.5 \times 173 + 16 \times 90 = 2564.5$$

Total area $= 17.5 \times 213 + 7 \times 215 + 20 \times 229 + 15 \times 229 + 15 \times 173 + 16 \times 90$

$$= 17282.5$$

$$\frac{2564.5}{17282.5} \times 8\,634\,750 = 1\,281\,285 \text{ people}$$

This is an estimation as the distribution of the data in the category 59 to 74 years is unknown and it assumes each age has an equal likelihood

13 c)

The data is not symmetrical around the mean, so it wouldn't be. Using the mean and standard deviation from part a), the normal distribution would be centred around 39.4 years. With a standard deviation of 23.1 the majority of the data would have to lie between -29.9 years and 108.7 years, which doesn't work as you can't have people less than 0 years old.

13 d)

The final category finishes at 90.5 which isn't a fair representation of people over the age of 90